# Payne (Pei En) Yeh

portfolio-payneyeh.vercel.app | github.com/Payne721401 | 90727sam@gmail.com | 0963-581-713

## Summary

Software engineer with a background in atmospheric science, applied ML, and data engineering. Experienced in building backend APIs, CI/CD automation pipelines, and embedded IoT systems. Passionate about bridging real-world data infrastructure with AI-powered applications.

## Work Experience

**Freelance Backend Developer** *Apr–Oct 2025*

Python | Flask | PostgreSQL | TDD

- Developed RESTful APIs in Python Flask following Test-Driven Development (TDD), achieving 100% test coverage across all endpoints.
- Designed PostgreSQL database schemas and built query indexes to optimize high-frequency read operations.

**Software Development & Cybersecurity Intern** *Mar–Jul 2024*
**Ernst & Young (EY) Consulting**

Python | Django | Selenium | SQL

- Architected an automated bidding document crawler and search system using Django MVC architecture.
- Optimized Python Selenium scraping scripts to process and structure 5,000+ external procurement records.

## Selected Projects

**Taiwan AI Weather App** *Mar–Oct 2025*

Flutter | Python | GitHub Actions | Firebase | Cloudflare R2 | Vertex AI

- Integrated Vertex AI (Gemini) to power an in-app AI weather assistant providing personalized forecasts and radar analysis.
- Implemented Geohash spatial indexing on Firestore, reducing database reads by ∼85% across 460+ weather stations.
- Downsampled forecast grids 1×1 km → 5×5 km (∼95% smaller payload); local caching cut API calls by ∼70%.
- Built GitHub Actions CI/CD pipelines for automated testing, data ingestion, and Telegram alerting.

**MQTT-based IoT Smart Classroom** *Feb–Aug 2023*

C/C++ | ESP32 | MQTT | Raspberry Pi | YOLOv8

- Developed ESP32 MCU firmware (C/C++) and a Raspberry Pi MQTT broker for real-time IoT sensor control and device automation.
- Integrated YOLOv8 for real-time indoor occupancy counting (95%+ accuracy) to automatically adjust HVAC and lighting.
- Awarded Finalist, National Smart Innovation Cross-disciplinary Competition (2023).

**HRV-based Sleep Stage Classification** *Feb–Jun 2024*

Python | Scikit-learn | PyTorch | Signal Processing

- Applied Empirical Mode Decomposition (EMD) on PhysioNet clinical dataset (1,985 subjects) to extract time-frequency HRV features.
- Trained Random Forest, Decision Tree, and SVM models for 5-class sleep stage prediction; achieved 82% accuracy on held-out test set.

## Education

**National Taiwan University**
MS in Atmospheric Sciences
Starting Sep 2026

**National Central University**
BS in Atmospheric Sciences
AI Application Credit Program
Sep 2020 – Jun 2024
Outstanding Student Scholarship (2024)

## Skills

- **Languages**: Python, C/C++, SQL, JavaScript/TypeScript, Dart
- **Backend**: Flask, Django, PostgreSQL, RESTful APIs
- **DevOps**: GitHub Actions, Docker, Linux, Firebase, Cloudflare R2
- **AI/ML**: PyTorch, Scikit-learn, Vertex AI (Gemini)

## Certifications

- Microsoft AZ-900 – Azure Fundamentals
- Microsoft AI-900 – Azure AI Fundamentals

## Awards & Competitions

- 1st Place, NCU Sustainable Innovation Proposal Competition (2023)
- 2nd Place, Weather Hackathon (2023)
- Finalist, National Smart Innovation Cross-disciplinary Competition (2023)

## Languages

- English: TOEIC 805